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Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
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Comments:			
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
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Task Description:	Highly energetic charged particles in space environments are hazardous because they cause DNA damage, which leads to mutation and loss of expression of critical genes that normally prevent cells from becoming cancerous. Most charged particle-induced mutations at autosomal loci span large regions of an affected chromosome resulting in loss of heterozygosity (LOH) for these regions. The proposed work has three interrelated goals. The first is to document the induction of mutational processes leading to LOH in two tissues, lung and spleen. The mutagenic effects of three charged particles will be tested: 1 GeV/amu Fe ions, 1 GeV/amu Ti ions, and 1 GeV protons to address the impact of ionization density on mutagenesis. The second goal is to demonstrate that LOH occurs on multiple chromosomes in the induced mutant cells, which would suggest the development of genomic instability. The third goal is to demonstrate directly with novel color markers that genomic instability characterized by high levels of LOH develops in the progeny of cells exposed in vivo. The basis for this approach is that selection of cells with LOH type mutations will provide the "bait" to		

	"fish out" cells exhibiting charged particle-induced genomic instability. Successful completion of the proposed work will identify those types of mutational events that reflect charged particle-induced genomic instability, and the role of ionization density in the development of these unstable cells in apparently normal tissues. The results will be useful in assessing cancer risks from space travel in two tissues that are prone to radiogenic cancers in humans.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Damage from ionizing radiation is both an environmental concern and a concern for cancer patients receiving radiation therapies, most notably for cancer. Our work provides information on how ionizing radiation can damage cellular genomes. In addition to showing that ionizing radiation creates damage at the target locus of interest, we have learned that cells suffering from one mutation are likely to have suffered from multiple mutations. This observation describes a subset of irradiated cells that might be at a higher risk to become cancerous. We also showed that for highly energetic large nucleus ions (e.g., iron and titanium), even a single particle passing through a nucleus may be sufficient to cause a large mutation event including chromosome translocations. The innovative technology we have developed is to first irradiate cells in a living organism, next isolate mutant cells from a tissue of interest, and then use cytogenetic approaches to examine the chromosome that has the mutational event. This approach gives up a very sensitive assay for large mutations caused by accelerated particles in living tissues.
Task Progress:	OVERVIEW The objective of this project was to determine the effects of accelerated particles found in the space environment on cells in intact organs. These particles are positively charged because they are nuclei stripped of their electrons, and they cause radiation damage when they move through cells within the organs. The three particles under study were iron (Fe) ions, titanium (Ti) ions, and protons (a hydrogen nucleus). Specifically, we examined the mutagenic effects from these accelerated particles. Mutations are changes in the DNA that often lead to changes in how genes function. Mutations are one of the causes of cancer formation. While other groups are studying the types of cancers caused by exposures to accelerated particles, we are determining how these exposures cause mutations to better understand how particle exposures cause cancer. In other words, we are studying the middleman between exposure and cancer. To determine the mutagenic effects in the intact organs, live mice were exposed to the accelerated particles at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). The mice were then returned to Oregon Health & Science University (OHSU) in Portland for determination of the frequencies and types of mutations caused by exposure to the particles. Radiation-induced mutations were measured in the cells from three tissues: lung, spleen, and kidney. Most data were obtained from the spleen, which contains white blood cells that end up in the blood, and kidney, which contains epithelial cells that filter toxins from the blood. The lung proved more difficult to work with because we did not get a sufficient number of mutant cells for analysis and will not be discussed further here. A particular focus over the last year was to determine whether low numbers of particle hits per cell would be sufficient to cause mutations because these numbers best mimic the space environment. The term used for the number of particles that hit a cell is “fluence”, and we use the term “low fluence” to describe space-like conditions under which cells are hit only a few times on average by particles. Mutations caused by radiation exposure often involve large regions of one or more chromosomes and we therefore determined whether mutant cells had mutations characteristic of radiation exposure. We term these “radiation signature mutations.” The radiation doses of most interest were low fluences where each cell was hit by only one or two accelerated Fe or Ti ions. These doses are below the level at which readily visible harm to cell can be observed. In other words, the doses tested are not sufficient to kill the cells. Our goal was to determine whether low fluences of Ti and Fe ions were sufficient to cause mutations because a surviving cell with one or more mutations can potentially become a cancer. The work is separated below to highlight the results for the two different cell types, kidney epithelial cells and spleen T cells. KIDNEY EPITHELIAL CELLS We first showed that relatively high fluences of Fe ions and Ti ions that, on average, struck each cell nucleus 8 times, would increase the frequency of mutant cells in the mouse kidney by about 3-4 fold. We also showed that these exposures caused mutations involving loss of large regions of the target chromosome. Many of these large mutations are “signature mutations” for radiation exposure because they are not found to occur in the absence of particle exposure. In some mutants, a portion of the target chromosome was lost and the remaining portion would fuse to another chromosome. This is termed a “chromosome translocation.” In the mutant cells, we also found that large regions of other chromosomes were affected frequently indicating widespread damage in the mutant cell’s genome. In total, the work with mutant kidney cells induced by high doses provide 4 markers for the radiation effect: 1) increased mutant frequencies, 2) signature mutations on the target chromosome, 3) mutations on other chromosomes, and 4) chromosome translocations. Having identified markers of mutagenic effect in mutant kidney cells from kidneys exposed to high fluences, we turned our attention to what happens at the low fluences of 1 or 2 particle hits per cell nucleus. For Ti ions, we observed all four markers at both fluences. For Fe ions at 2 particles per cell nucleus, we observed increased mutant frequencies, increased radiation signature mutations, and increased mutations on other chromosomes. (We did not measure translocations for the mutant cells obtained from the Fe ion exposed kidneys.) For Fe ions at a fluence of 1 particle per cell nucleus, we observed increased mutant frequencies and increased mutations on other chromosomes, but no increase in radiation signature mutations. Considered as a whole, our work with kidney epithelial cells showed the induction of mutations even in cells that on average were only struck one time by charged Ti and Fe particles. These results show that for Ti and Fe particles, on a per cell basis there really is no minimum exposure level at which a mutagenic effect will not occur. Because mutants cause cancer and epithelial cancers are induced by charged particles, our work suggests that the mechanism of charged-particle cancer induction is via the induction of mutations. Finally, work with high doses of protons (a proton is an accelerated hydrogen nucleus) has clearly demonstrated the radiation signature markers discussed above. However, work at low doses to date has failed to demonstrate these markers, which suggests that the small hydrogen nucleus cannot cause as much damage per particle as the larger Ti and Fe ions. SPLEEN T CELLS Cancers of the white blood cells can be leukemias or lymphomas. Blood cell cancers induced by radiation exposure

	appear much sooner after exposure than solid tissue cancers, suggesting that differences exist between epithelial and blood cell types. We determined the frequencies and types of mutations in T cells in the same set of mice discussed above, and observed both differences and similarities compared with the kidney cells. The first difference was that we did not observe consistent increases in T cell mutant frequencies regardless of the type of particle or dose to which the mice were exposed. The second difference was that we did not observe radiation signature mutations in the spleen cell mutants at the highest doses tested. However, like the kidney cells we did observe radiation signature mutations at the two low doses tested (average 1 or 2 particle hits per nucleus for Fe and Ti ions) despite the absence of these mutations at the higher doses. This observation at the low fluences demonstrates that like the kidney cells, the spleen T cells provide a very sensitive assay for the types of mutations induced by accelerated particles present in the space environment. Due to the limited growth of the spleen T cells, we were unable to look for chromosome translocations in the radiation-induced mutants. During the next funding period (with new grant NNX14AC97G), one goal is to develop a new assay to do this analysis. Finally, we believe the difference between the kidney epithelial cells and spleen T cells with regard to the presence or absence of radiation signature mutations after exposures to high doses of the accelerated particles, respectively, is due to differences in sensitivity and/or turnover. Kidney cells are more resistant to radiation exposure and they turnover infrequently, in comparison to spleen T cells, which are sensitive to radiation and turnover relatively frequently. While interesting, the differences are not as important as the demonstration that mutations are induced in both cell types at low fluence because these conditions exist in the space environment and will pose risks to astronauts.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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